



INTERNATIONAL LEVEL HACKATHON 360° – 4.0 • ROUND 1

Narnarayan Shastri Institute of Technology & ECLearnix EdTech Pvt. Ltd. Organizes

TEAM DETAILS

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Domain	Domain 1 – Open Challenge Solutions
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SL. No	Team Member Name	Hackathon ID	WhatsApp Number	Role	Designation	College / Dept / State / Country
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Problem Statement



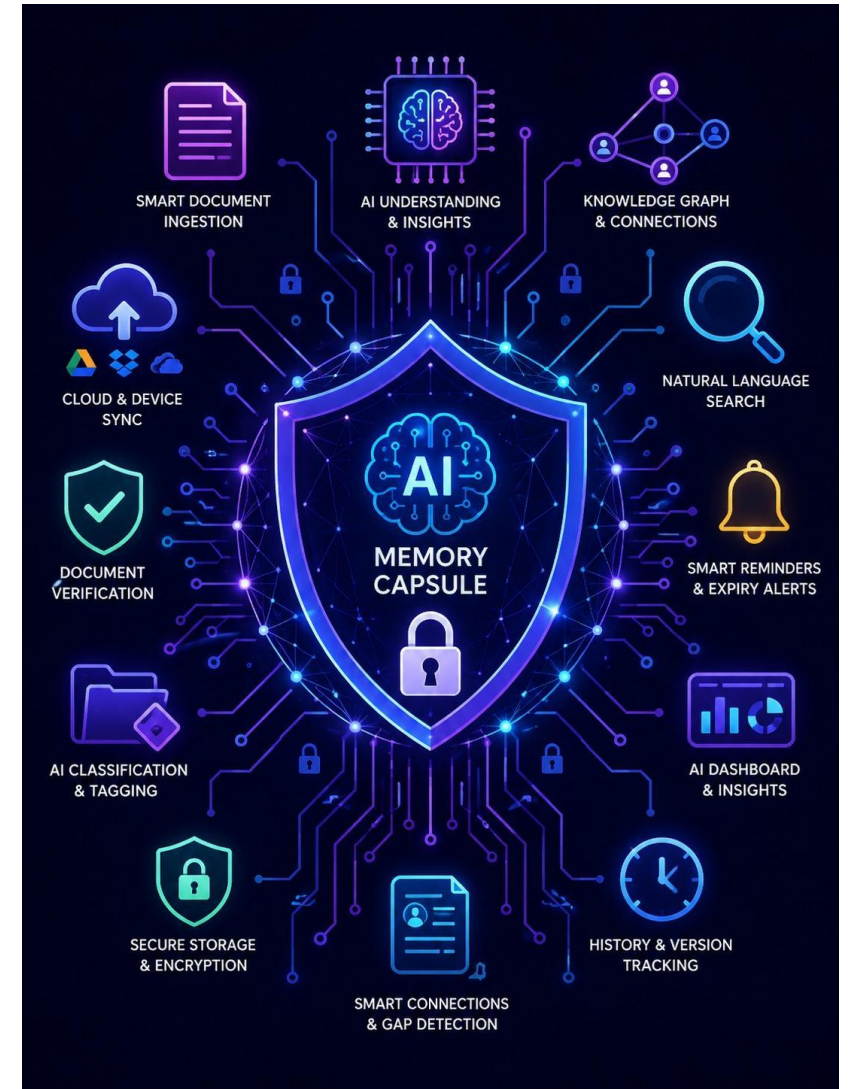
The Challenge

Individuals' digital documents are scattered across multiple platforms, making them difficult to organize, search, verify, and manage. Existing storage solutions lack AI-powered understanding, intelligent search, and smart insights.

DOMAIN 1 – OPEN CHALLENGE SOLUTIONS

AI Memory Capsule

An intelligent document intelligence platform that transforms scattered personal records into a searchable, verifiable, and interconnected knowledge graph with AI-powered insights, reminders, and relationship discovery.



Project Objectives

This system is built around five core objectives to create an AI-powered personal document intelligence platform.

1

Smart Data Collection

Securely collect and store personal documents from multiple sources.

2

AI Classification & Understanding

Automatically classify documents and extract key information using AI.

3

Knowledge Graph & Search

Connect related documents and enable natural language search.

4

Verification & Smart Alerts

Verify document authenticity and provide expiry reminders.

5

Dashboard & Insights

Provide a centralized dashboard with history, smart recommendations, and document analytics.

Existing System Overview

1

Fragmented Storage Across Platforms

Personal documents remain scattered across cloud drives, email attachments, phone galleries, and physical folders with no unified structure.

2

Manual Search & Organization

Users must remember file and folder names to locate documents, with no AI-based classification or natural language search available.

3

No Verification or Authenticity Checks

Existing tools only store a copy of the document; they cannot confirm authenticity or detect tampering, and documents stay disconnected from one another.

4

No Smart Alerts or Insights

No tracking of certificate expiry dates, no proactive reminders, and no centralized dashboard connecting related records or insights.

Proposed System Overview

1

Smart Document Collection

Securely collect documents from PDFs, images, cloud storage, emails, and mobile devices into a centralized digital archive.

2

AI Classification & Understanding

Automatically classify documents and extract important information using OCR and AI to understand their content.

3

Knowledge Graph & Natural Language Search

Connect related documents, people, organizations, and events into an intelligent knowledge graph, allowing users to search using natural language.

4

Intelligent Verification & Smart Insights

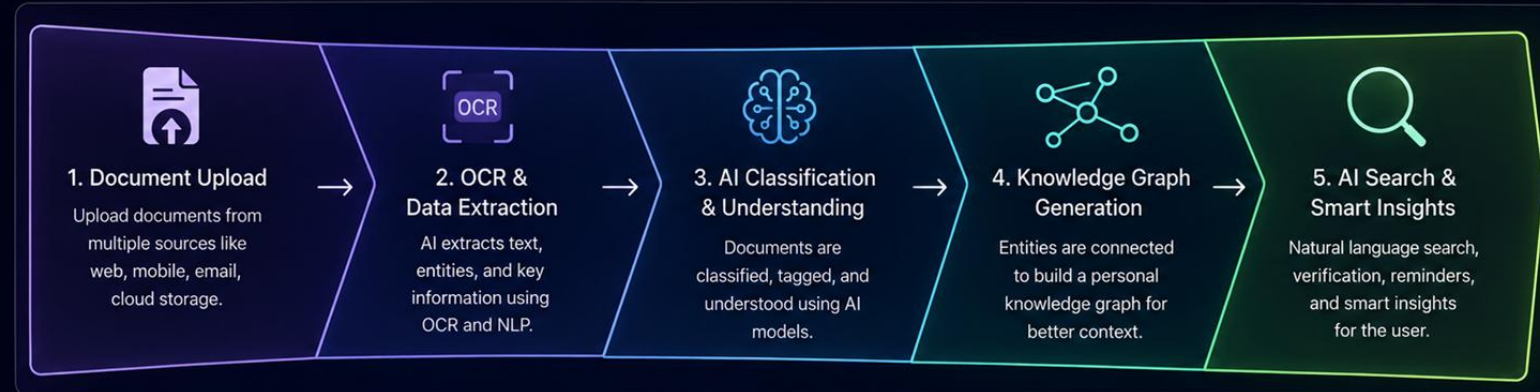
Verify document authenticity using QR codes, digital signatures, and AI-based tamper detection while providing expiry reminders, document recommendations, and a personalized dashboard.



AI Methodology

AI Methodology: How AI Memory Capsule Works

From Documents to Intelligence



AI Processing Examples

	Upload Degree Certificate	→	Extract Name, University, Year	→	Education Category
	Upload Medical Report	→	Extract Hospital, Diagnosis, Date	→	Medical Category
	Upload Income Certificate	→	Detect Expiry & Certificate Number	→	Verification Ready
	Upload Internship Certificate	→	Connect to Resume & Skills	→	Knowledge Graph Updated

AI Output Sample

Document Type: Education Certificate
Category: Education ✓

Extracted Information:

- Name: Midhun M
- University: APJ Abdul Kalam Technological University
- Year: 2026

AI Insights:

- ✓ Related to B.Tech Degree
- ✓ Added to Education Timeline
- ✓ Available through Natural Language Search
- ✓ Verification Status: **Verified**

Feature Modules

1

Smart Document Collection

Securely upload documents from PDFs, images, emails, cloud storage, and mobile devices. AI automatically extracts text and metadata using OCR.

2

Natural Language Search

Search documents using simple questions like “Show my internship certificates” or “Find my insurance policy” and receive accurate AI-powered results.

3

AI Classification & Understanding

Automatically classify documents into categories such as Education, Identity, Finance, Medical, Insurance, Projects, and Achievements while extracting key information.

4

Document Verification

Verify authenticity using QR codes and certificate numbers.

5

Knowledge Graph & Rewind Timeline

Connect documents, people, organizations, events, and achievements into an intelligent knowledge graph with a visual life timeline.

6

Smart Remind

Track expiry dates and send timely notifications.

Feature Modules

7

AI Dashboard

View document status, insights, analytics, and recommendations.

8

Secure Storage

Protect documents with encryption and secure access.

9

Smart Connections

Identify missing documents and discover relationships between records.

10

History Tracking

Maintain document versions and activity history securely.

What Sets Us Apart

1

Unified Knowledge Graph, Not Just Storage

Goes beyond folders and files by linking documents, people, organizations, and events into one connected graph instead of isolated storage.

2

Natural Language, Not Keyword Search

Users ask plain-language questions like “Show my internship certificates” instead of navigating folders or remembering exact file names.

3

Built-In Verification & Tamper Detection

Combines QR codes, digital signatures, and AI-based tamper detection to confirm authenticity, not just store a copy of the document.

4

Proactive, Not Reactive

Tracks expiry dates and missing documents on its own, alerting users before a certificate lapses instead of waiting to be searched for.

5

Risk-Aware Security From Sign-Up

Assigns a dynamic risk score during registration and login to catch suspicious activity early, alongside encryption and multi-layer authentication.

6

A Career Companion, Not Just an Archive

Builds a professional portfolio from stored achievements and recommends skills and opportunities, turning stored documents into career guidance.

Results & Observations

System behavior was validated across representative document management scenarios, demonstrating accurate AI classification, intelligent retrieval, and reliable verification.

Test Case	Observed Signal	System Response
AI Document Processing	Uploaded document analyzed	Information extracted, categorized, and securely stored.
Smart Search	User searches using plain language	Relevant document retrieved instantly using AI semantic search.
Verification	QR / Certificate ID detected	Document verified.
Expiry Alert	Expiry date detected	Reminder generated.

Results & Observations

Test Case	Observed Signal	System Response
Knowledge Graph	Related documents found	Connections created.
AI Suggestions	Missing document detected	AI recommends upload.
History	Document updated	Activity logged.

Expected Impact & Benefits

1

Intelligent Document Management

Automatically organizes personal documents into meaningful categories, reducing manual effort and making information easier to access.

2

Career & Skill Development

Helps users build a comprehensive professional portfolio by tracking achievements, suggesting skill improvements, and recommending relevant career opportunities.

3

Improved Productivity

Enables instant retrieval of documents and information through natural language search, saving time spent searching across multiple platforms.

4

Smart Insights & Reminders

Provides proactive reminders for certificate expirations, document renewals, and learning milestones, ensuring users never miss important deadlines.

5

Secure Personal Data Management

Protects sensitive information using encryption, secure authentication, and user-controlled access while maintaining privacy.

6

Trusted Credential Management

Supports professional verification by organizing certificates, maintaining their validity status, and integrating with trusted issuer verification systems where available.

Future Enhancements

1

Phase 1

Multi-source integration with cloud storage, email, and learning platforms.

2

Phase 2

Advanced AI for document understanding, summarization, and multilingual support.

3

Phase 3

Issuer-based certificate verification and secure digital identity management.

4

Phase 4

AI-powered career guidance with personalized learning and skill recommendations.

5

Phase 5

Cross-platform AI assistant with voice support, offline access, and proactive reminders.

Challenges



1

Intelligent Document Classification

Accurately categorizing diverse personal documents despite different formats and structures.

2

Data Privacy & Security

Protecting sensitive user data through encryption, secure storage, and permission-based access.

3

Credential Verification

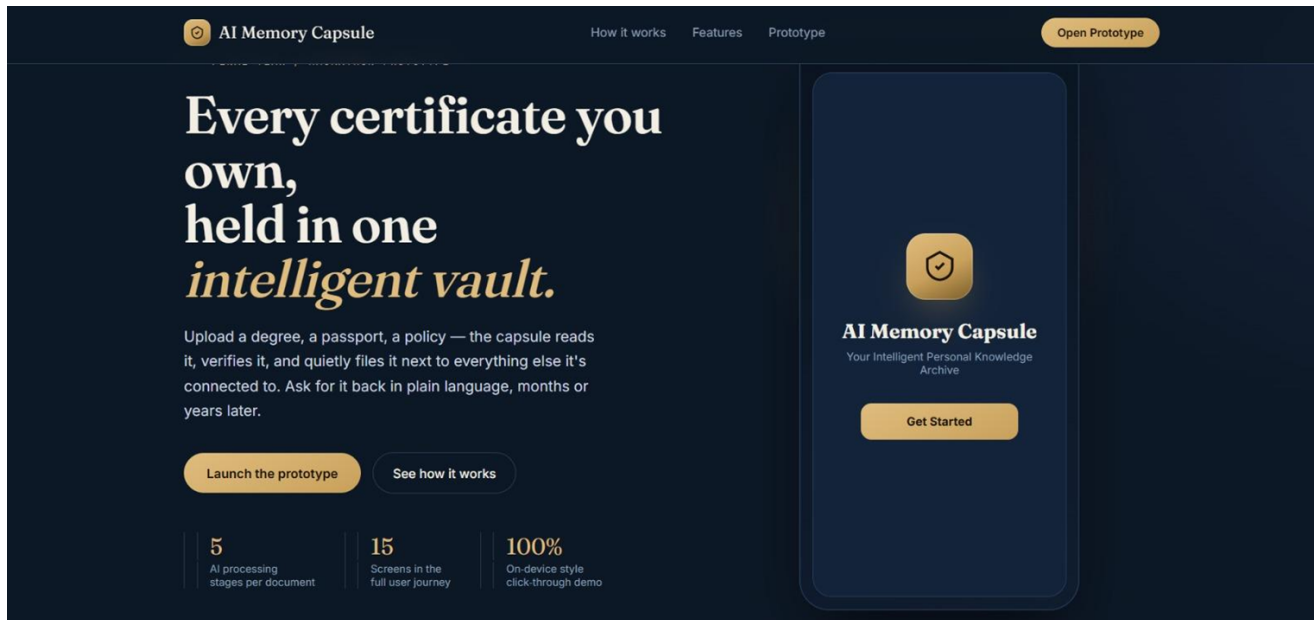
Validating certificates using trusted issuer databases while handling unverifiable documents transparently.

4

Scalable Knowledge Organization

Efficiently managing and connecting thousands of documents into an intelligent knowledge graph for fast retrieval.

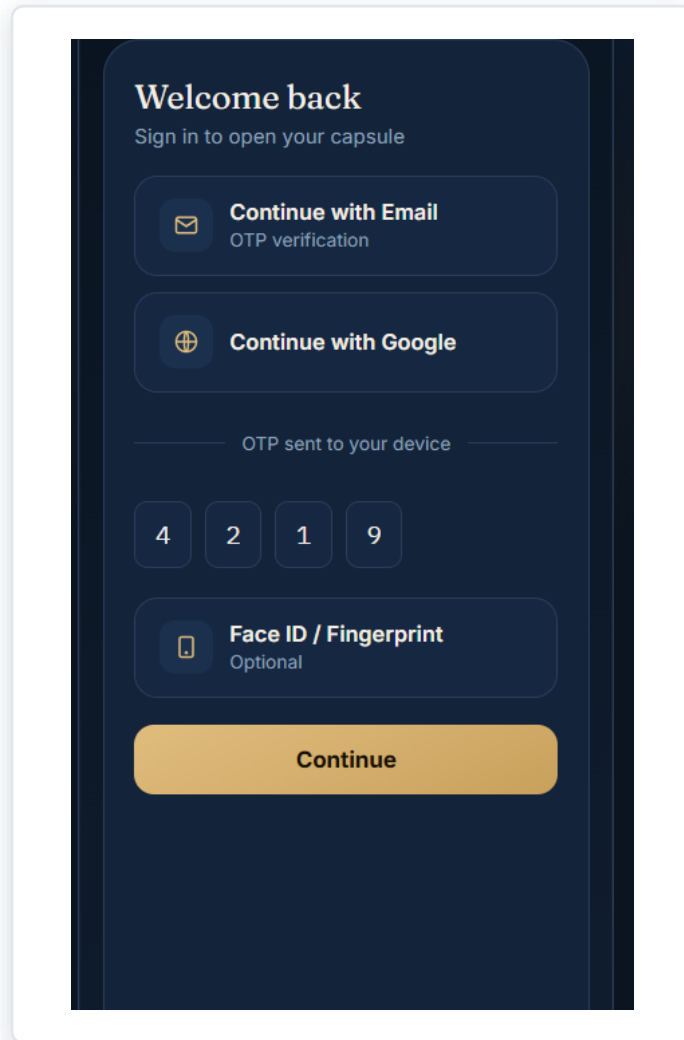
Login Page



ACCESS & SECURITY

Access your AI Memory Capsule using Password Login or Face Login. Intelligent security detects suspicious activities, calculates risk scores in real time, and protects your personal archive with multi-layer authentication.

Register Page



AI-assisted secure registration with OTP verification and Government ID upload.

- Collects and validates user identity information securely.
- Verifies the user's email or mobile number using OTP authentication.
- Supports Government ID upload (Aadhaar, Passport, Driving License, etc.) for identity verification.
- Uses AI to detect suspicious registrations such as fake emails, duplicate accounts, and bot activity.
- Assigns a dynamic risk score during registration to prevent fraud and ensure trusted onboarding.
- Encrypts all personal information and uploaded documents for maximum security.



Thank You

AI Memory Capsule for EC Learnix | Domain 1: Open Challenge Solutions

OPEN FOR QUESTIONS & DISCUSSION